

Cryptographic Algorithm Taxonomy

- Classified by Algorithmic Goals (Algorithms may appear in multiple categories)
 - Each category includes mathematical foundations
 - SM algorithms integrated naturally (Chinese standard)
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1. Asymmetric Encryption Algorithms

Goal: Public-key encryption / key transport **Math Foundations:**

- Integer factorization (RSA)
- Finite-field discrete logarithm (ElGamal)
- Elliptic curve discrete logarithm (ECIES, SM2)

Algorithms:

- RSA (OAEP, PKCS#1 encryption)
 - ElGamal Encryption
 - ECIES
 - SM2 Encryption (Chinese standard, ECC-based)
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2. Digital Signature Algorithms

Goal: Authentication, integrity, non-repudiation **Math Foundations:**

- Integer factorization (RSA signatures)
- Finite-field discrete logarithm (DSA)
- Elliptic curve discrete logarithm (ECDSA, EdDSA, SM2)

Algorithms:

- RSA Signatures (PSS, PKCS#1 v1.5)
 - DSA
 - ECDSA
 - Ed25519 / Ed448
 - SM2 Signature (Chinese standard)
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3. Key Exchange / Key Agreement Algorithms

Goal: Derive shared secrets over insecure channels **Math Foundations:**

- Finite-field discrete logarithm (DH, DHIES)
- Elliptic curve discrete logarithm (ECDH, ECMQV, SM2)
- Montgomery curves (X25519, X448)

Algorithms:

- Diffie–Hellman (DH)
- DHIES

- ECDH
 - ECMQV
 - X25519 / X448
 - SM2 Key Exchange (Chinese standard)
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4. Symmetric Block Ciphers

Goal: Confidentiality via fixed-size block encryption **Math Foundations:**

- Substitution–permutation networks (AES, Camellia)
- Feistel networks (DES, Blowfish, Twofish, SM4)

Algorithms:

- AES (CBC, GCM, CTR, CFB, OFB)
 - DES / 3DES
 - Camellia
 - Twofish
 - Blowfish
 - IDEA
 - SM4 (Chinese standard)
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5. Symmetric Stream Ciphers

Goal: High-speed stream encryption **Math Foundations:**

- LFSR-based designs
- ARX constructions (Add–Rotate–XOR)

Algorithms:

- RC4
 - Salsa20
 - ChaCha20
 - HC-128 / HC-256
 - Grain family
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6. Cryptographic Hash Functions

Goal: One-way mapping, integrity, fingerprinting **Math Foundations:**

- Merkle–Damgård (SHA-1, SHA-2, MD5, RIPEMD)
- Sponge construction (SHA-3 / Keccak)

Algorithms:

- SHA-1
- SHA-2 (SHA-256 / SHA-512)
- SHA-3 (Keccak)
- MD5

- RIPEMD160
 - Whirlpool
 - SM3 (Chinese standard)
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7. Message Authentication Codes (MAC)

Goal: Integrity + authenticity with shared key **Math Foundations:**

- Hash-based MAC (HMAC)
- Block-cipher-based MAC (CMAC, GMAC)

Algorithms:

- HMAC (SHA-1 / SHA-2 / SHA-3 / SM3)
 - CMAC (AES, SM4)
 - GMAC (GCM authentication)
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8. Key Derivation Functions (KDF)

Goal: Derive strong keys from passwords or shared secrets **Math Foundations:**

- Iterated hashing
- HMAC-based PRF
- Memory-hard functions

Algorithms:

- PBKDF2
 - HKDF
 - KDF1 / KDF2
 - Scrypt
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9. Random Number Generators (RNG / DRBG)

Goal: Generate cryptographically secure random values **Math Foundations:**

- Entropy extraction
- Hash-DRBG
- HMAC-DRBG
- CTR-DRBG

Mechanisms:

- SecureRandom
 - SP800-90A DRBG family
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Notes

- Algorithms may appear in multiple categories (e.g., RSA, SM2).

- SM algorithms are integrated into their natural categories with a simple "(Chinese standard)" tag.
 - This taxonomy reflects real cryptographic usage rather than forcing uniqueness.
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